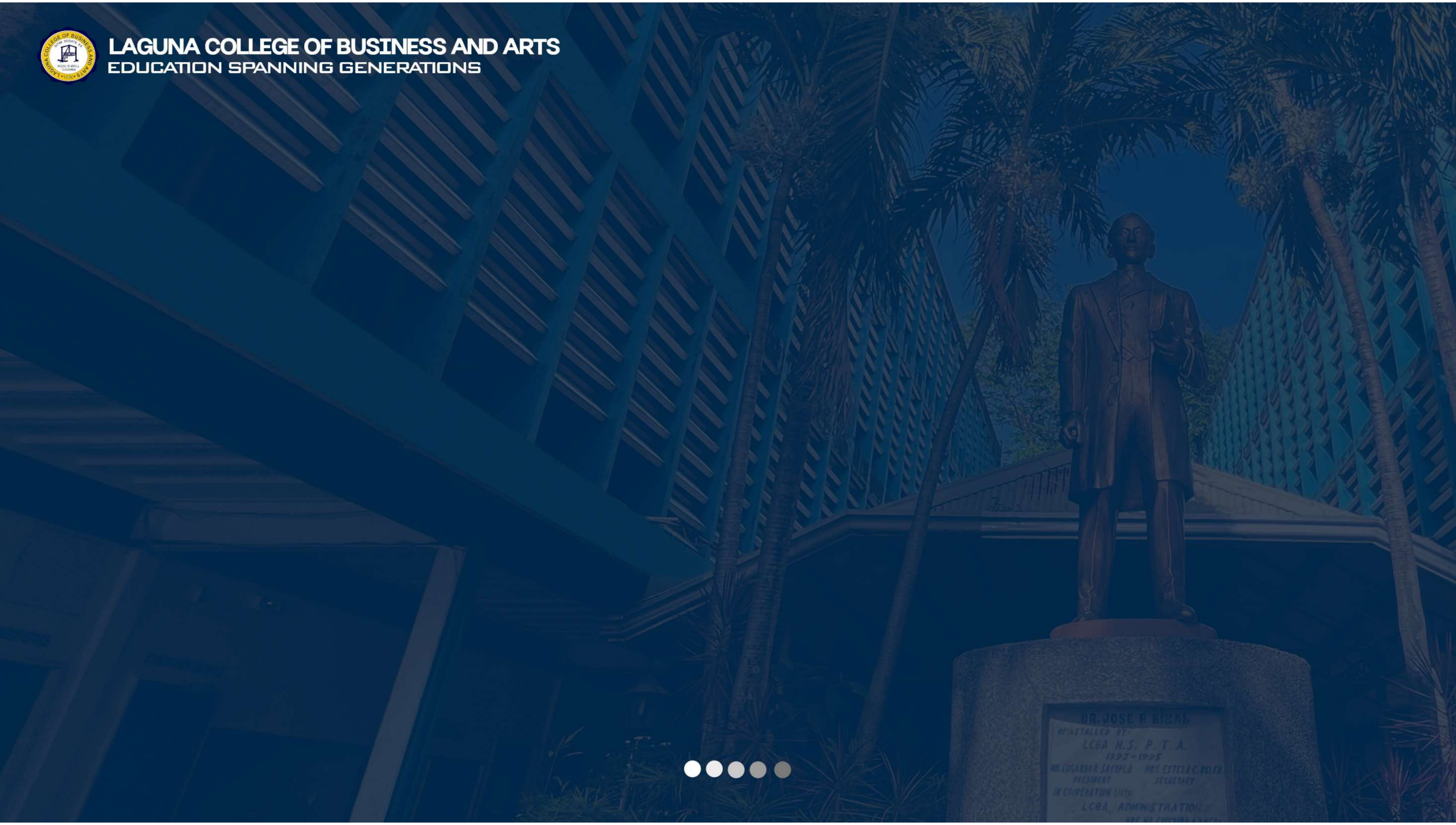
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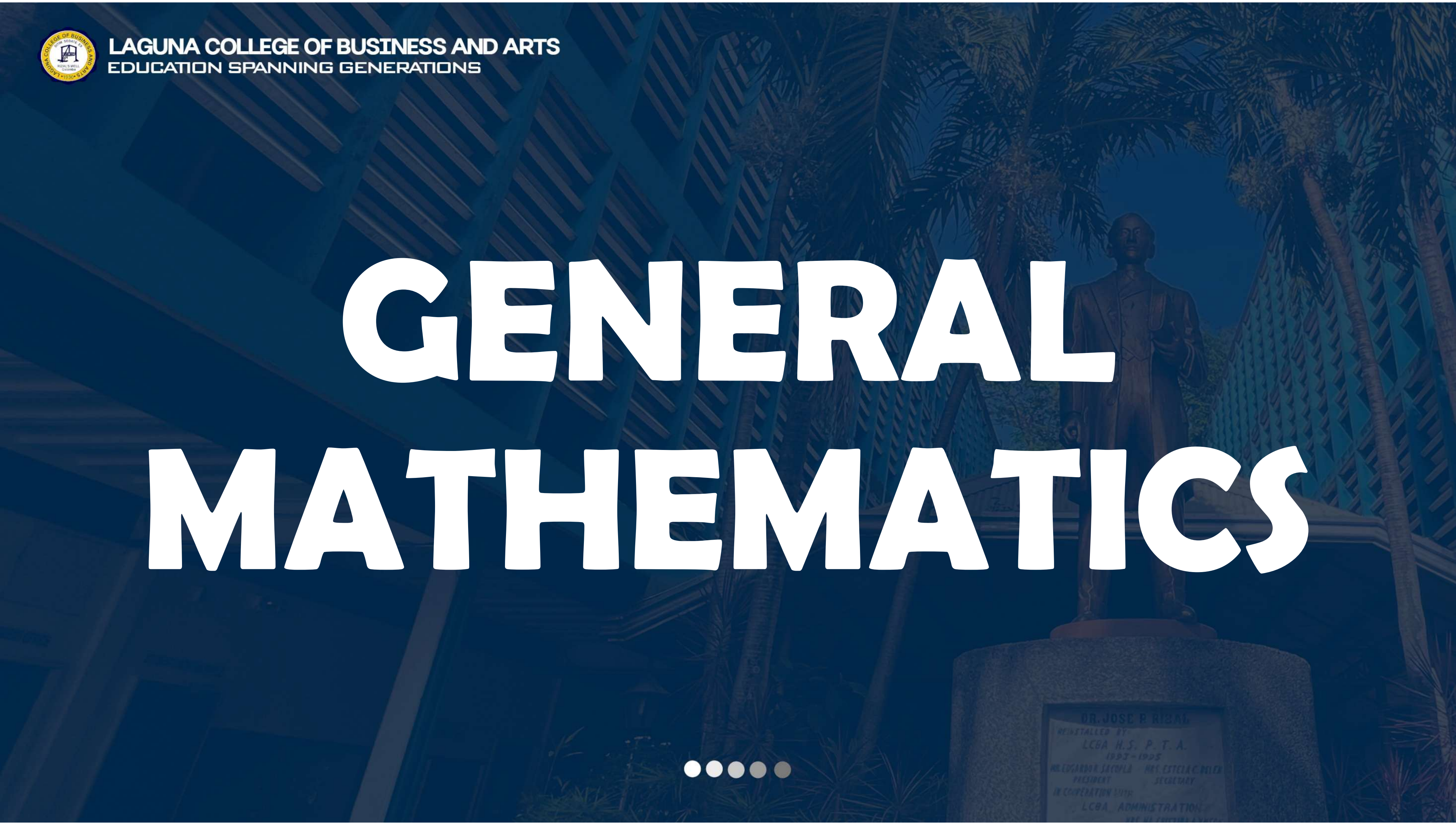
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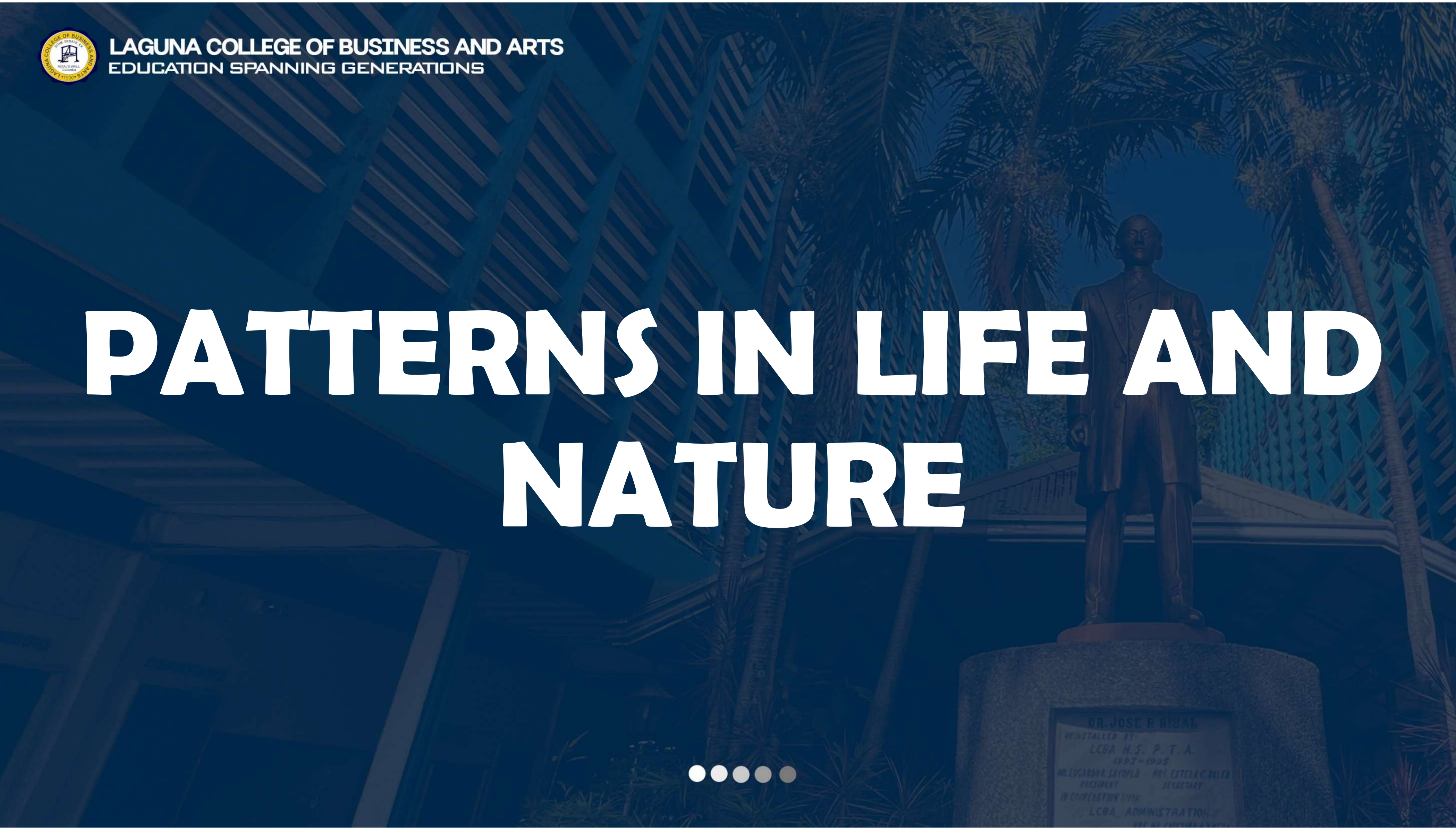
GENERAL
MATHEMATICS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PATTERNS IN LIFE AND NATURE

Learning Objectives:

1. define patterns, including special types such as the Fibonacci sequence, and identify patterns found in art, nature, and everyday life.
2. determine the next terms of a given pattern, formulate its governing rule, and analyze its significance in nature, human design, and mathematical reasoning.
3. create a visual showcase of at least two real-life patterns (through drawings, photographs, or digital examples) and present the rule or sequence behind each pattern.
4. appreciate the beauty, order, and importance of patterns in mathematics, nature, art, and everyday life by actively participating in class discussions and activities.

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ACTIVITY: PATTERN DETECTIVES: FINDING THE RULE

Directions: Study each pattern carefully. Fill out the table with the next three terms and the basis for determining them.

Sequence	Next 3 terms	What is your basis in determining the next 3 terms?
2, 4, 6, 8, …		
3, 6, 12, 24, …		
1, 1, 2, 3, 5, …		
$\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$		
1, 4, 9, 16, …		
6, 6, 6, 6, …		

Sequence	Next 3 terms	What is your basis in determining the next 3 terms?
2, 4, 6, 8, …	10, 12, 14	Each term increases by 2 (add 2).
3, 6, 12, 24, …	48, 96, 192	Each term is multiplied by 2.
1, 1, 2, 3, 5, …	8, 13, 21	Each term is the sum of the two previous terms (Fibonacci sequence).
1/2, 1/3, 1/4, 1/5, …	1/6, 1/7, 1/8	Denominator increases by 1 each time.
1, 4, 9, 16, …	25, 36, 49	Each term is a perfect square (n^2).
6, 6, 6, 6, …	6, 6, 6	All terms are constant (always 6).

PRE-ASSESSMENT

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Test I. Multiple Choice.

Directions: Choose the letter of the correct answer. Write your answer in CAPITAL LETTERS on the answer sheet.

1. Patterns in art and nature can be described by:

A. Guesswork

B. Careful inspection

C. Random selection

D. Trial and error

2. A pattern of petals in a flower is an example of:

A. Random sequence

B. Arithmetic sequence

C. Geometric sequence

D. Fibonacci sequence

3. The Fibonacci sequence starts with:

A. 0 and 1

B. 1 and 1

C. 2 and 3

D. 1 and 2

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4. In the Fibonacci sequence, the next term after 21 is:

A. 22

B. 32

C. 34

D. 36

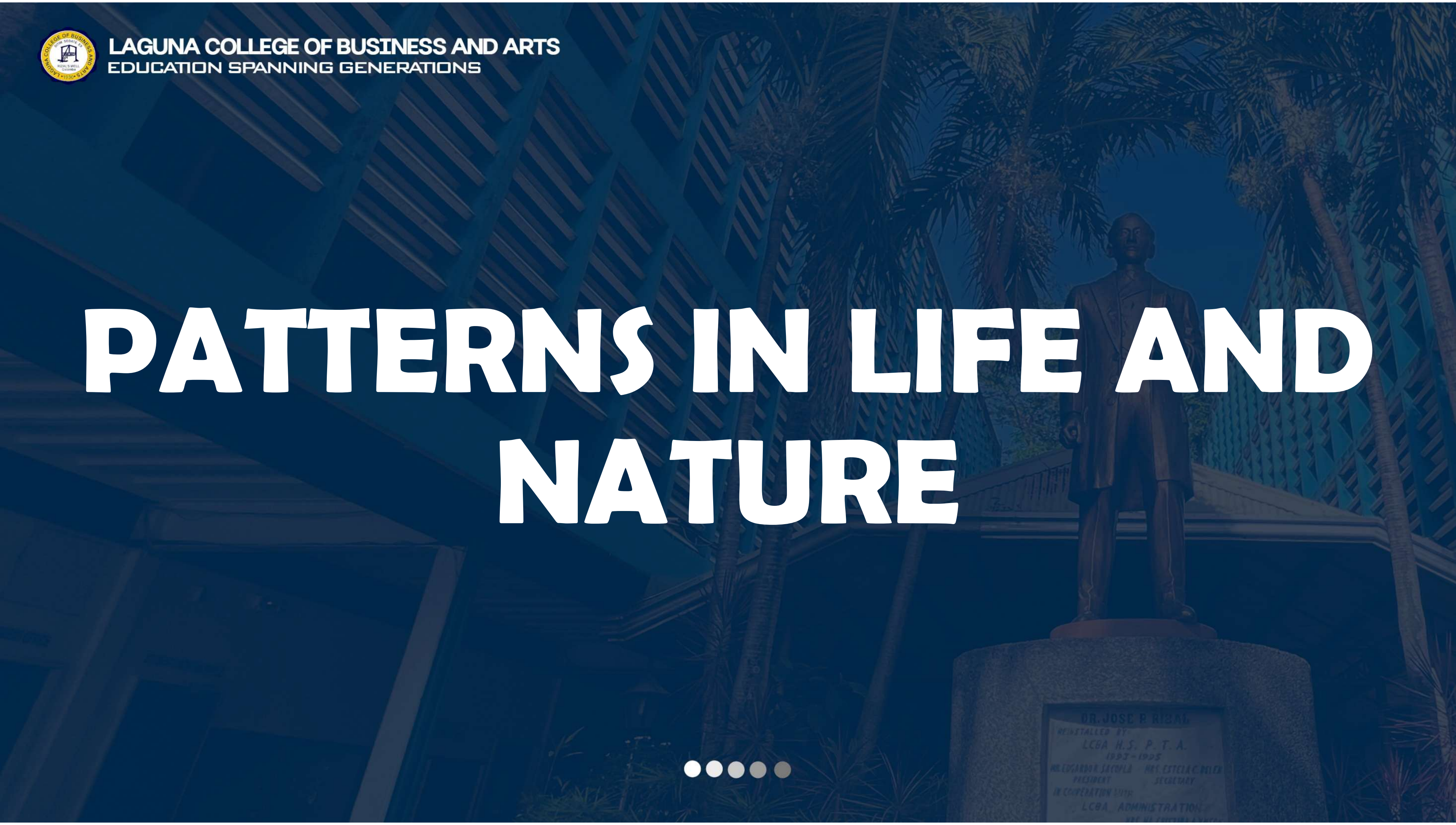
5. The rule of a pattern explains:

A. How each term is obtained

B. The sum of all terms

C. The random order of terms

D. The largest term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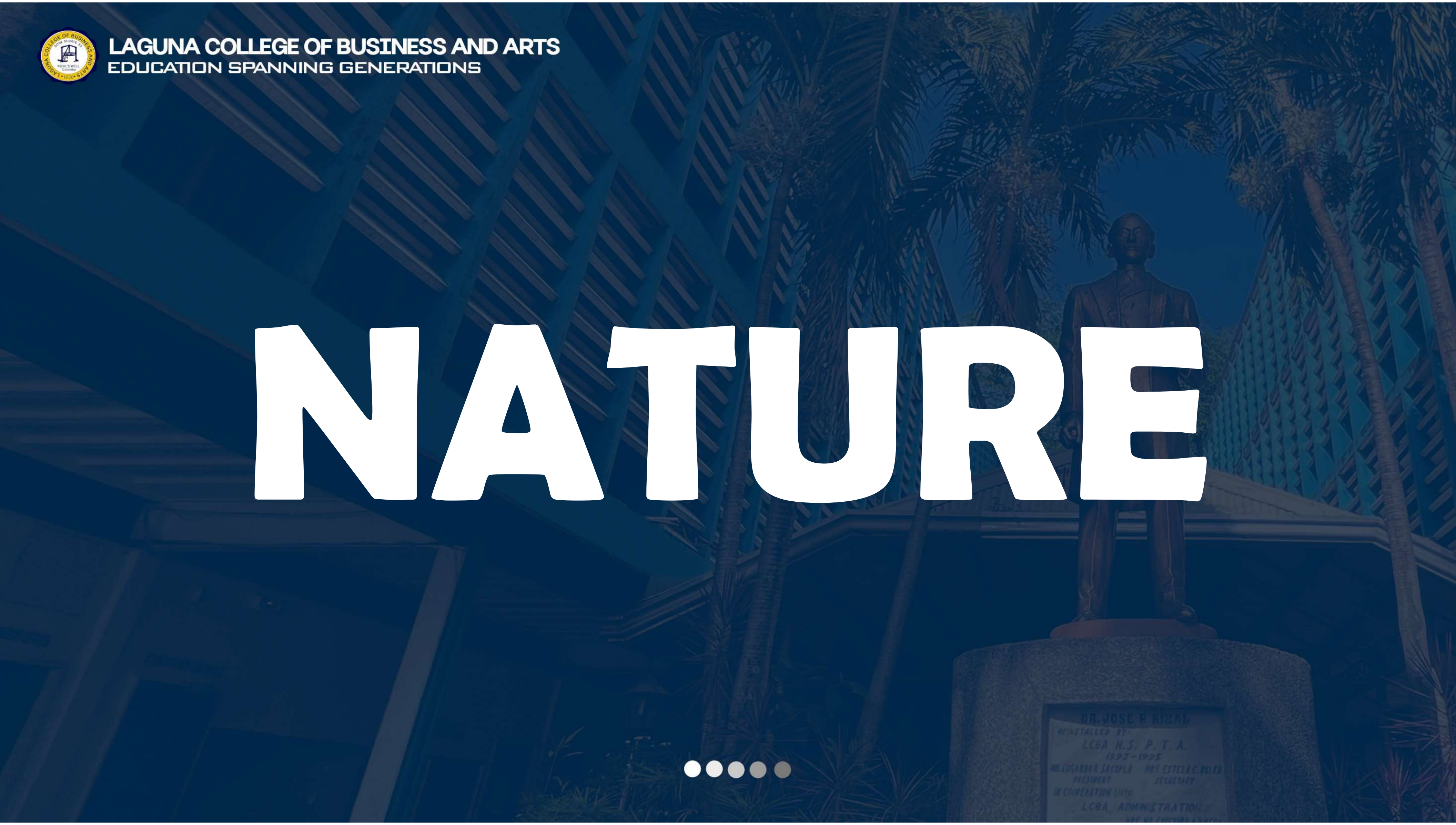
PATTERNS IN LIFE AND NATURE

Patterns Around Us

A pattern is a repeated or regular arrangement of shapes, numbers, or events. Patterns can be visual, like the spirals in shells or the symmetry in flowers, or numerical, like sequences of numbers that grow in a predictable way.

Patterns in nature often emerge from physical laws and biological principles such as growth, efficiency, and survival.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NATURE

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- **Fractals:** Found in trees, river networks, lightning bolts, and blood vessels; they model complex structures through repeating self similar forms.

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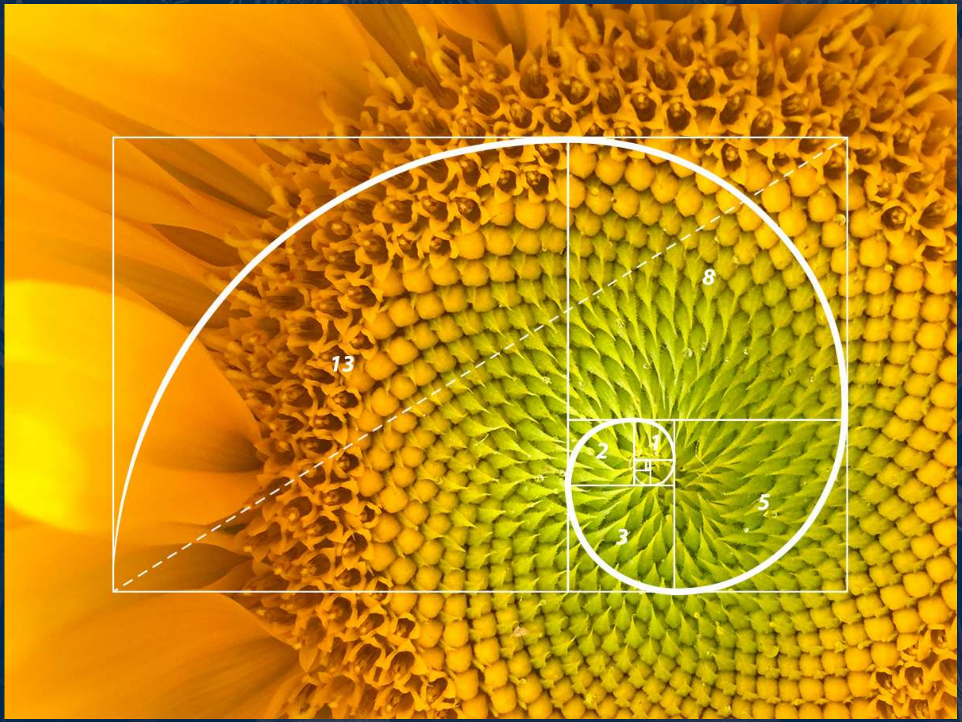
- **Fibonacci Sequence & Golden Ratio:** Seen in sunflowers, pinecones, nautilus shells, and hurricanes; these patterns optimize packing and growth.



- **Symmetry:**

Animals and plants often exhibit bilateral or radial symmetry, which can help in movement, camouflage, or mating rituals.

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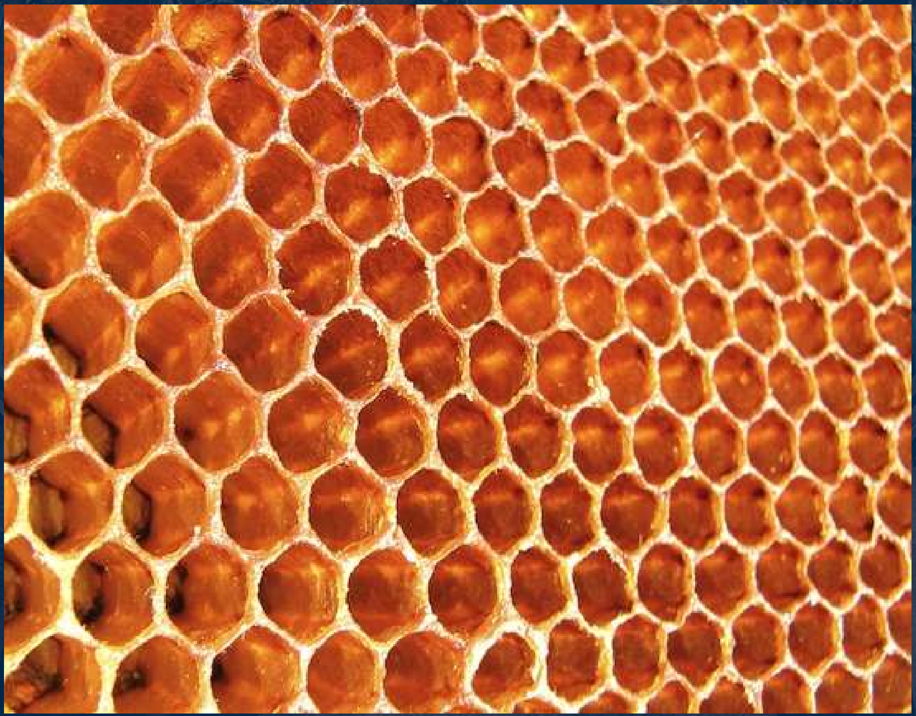
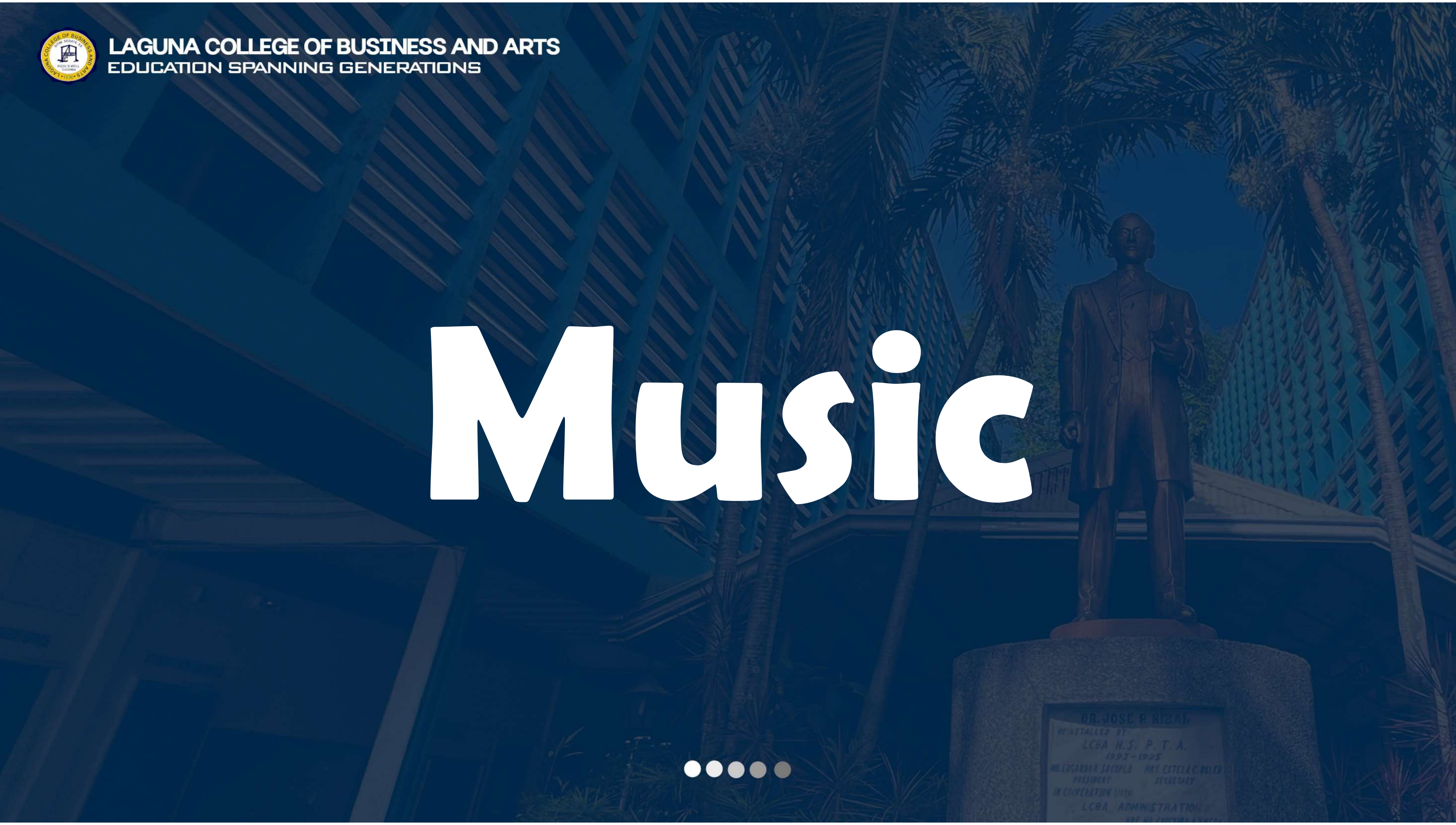
- **Tessellations and Spirals:** **Beehives, spider webs, and shells showcase geometric efficiency in nature.**

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Music

- **Rhythmic Patterns**

Repetition and variation in beats give music its pulse and emotional arc.

- **Scales and Chord Progressions**

Patterns within scales (e.g., major, minor, pentatonic) are foundational to musical language across cultures.

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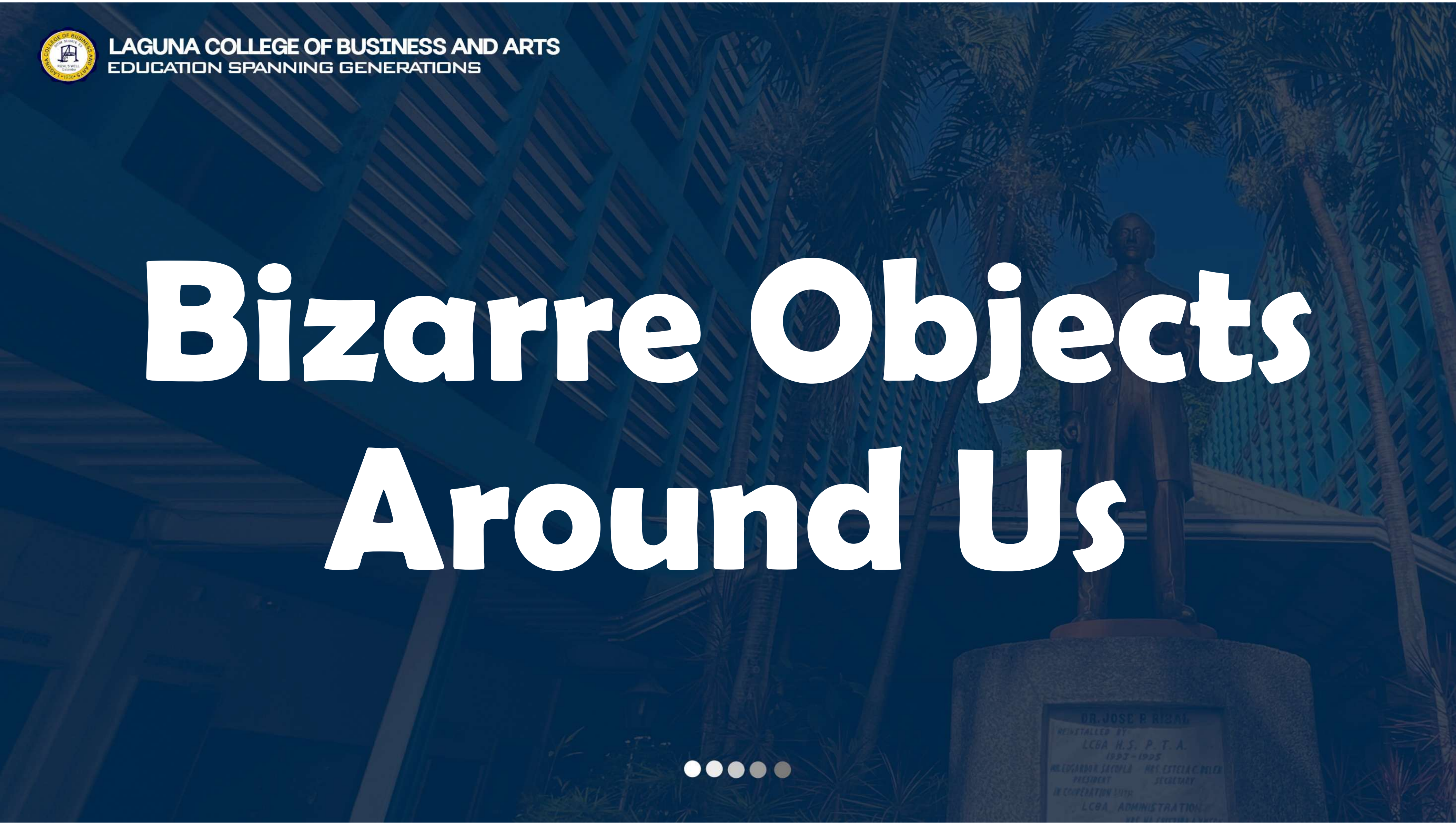
- **Mathematical Sequences**

Composers use Fibonacci numbers, fractals, or symmetry in composition (e.g., Bartók, Xenakis).

- **Repetition and Motifs**

Leitmotifs in operas or recurring riffs in pop music rely on patterns for emotional resonance.



Bizarre Objects Around Us

Cracked Mud or Paint

This form crazing or desiccation patterns, studied in geology and materials science.

Coffee Ring Effects

Evaporation leaves ring patterns that are now studied for drug delivery and inkjet printing.

Bubble Clusters and Foam

Packing and tessellation principles apply in soap films and cellular structures.

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Digital Glitches and Data Corruption

Even errors can produce repeating or pseudo-random patterns, useful in glitch art or data visualization.

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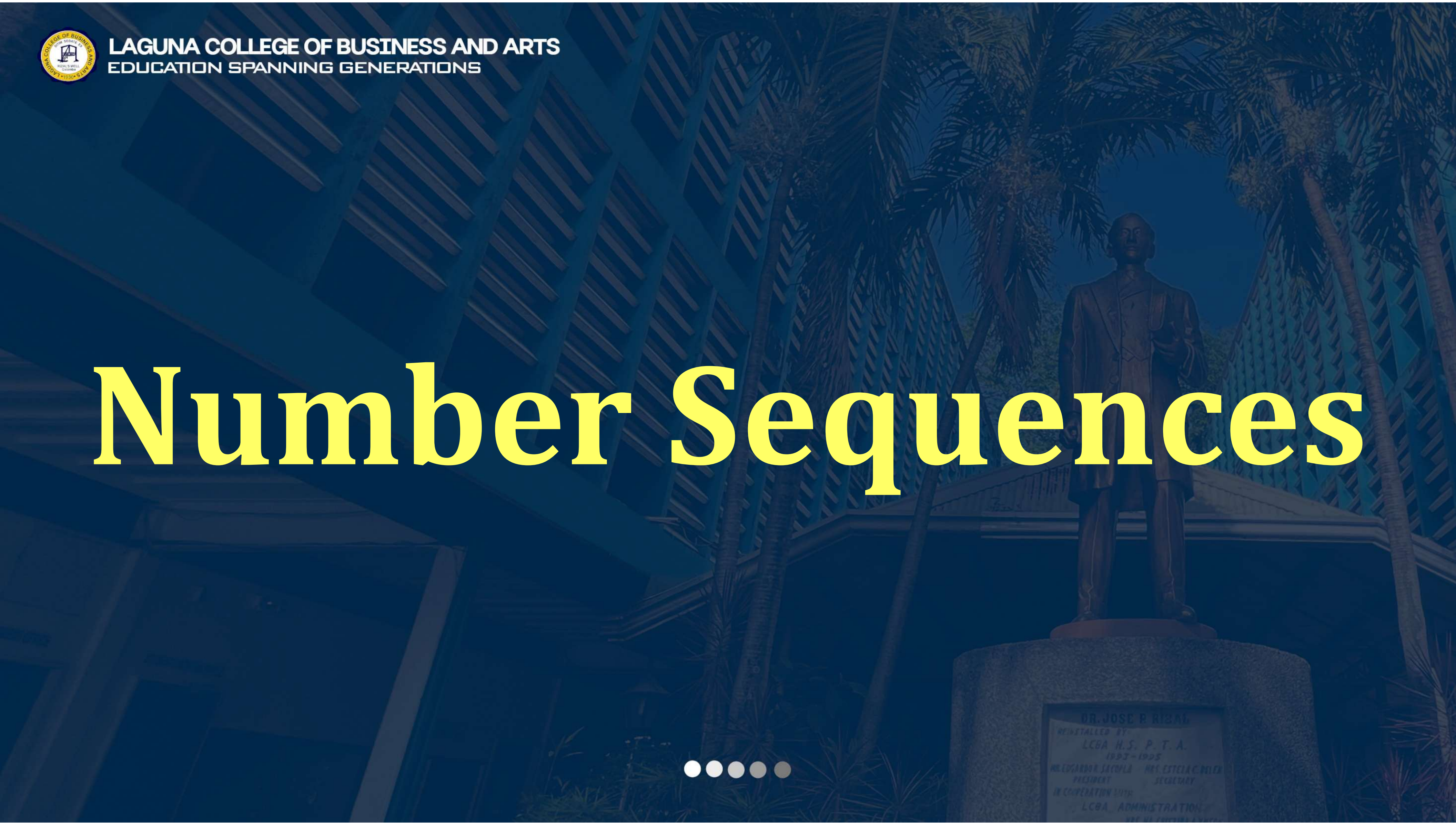

Number Sequences

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Number Sequences

A sequence is an ordered list of numbers that follow a certain rule. The rule explains how to get the next term.

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Types of Sequences

Arithmetic Sequence

is a sequence in which the difference between any two consecutive terms is the same.
It is also known as an arithmetic progression.

This constant difference is called the common difference, and it will be denoted by d.

Each term is obtained by adding the common difference, d.

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Arithmetic Sequence

It can be represented by the general formula:

$$a_n = a_1 + (n - 1)d$$

Where,

a_n= is the last term

a_1= is the first term

n = is the number of terms

d= is the common difference

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Illustrative Example 1:

Sequence: 2, 4, 6, 8, …

Common difference:

$$d = 2$$

Next three terms: 10, 12, 14

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Geometric Sequence

If arithmetic sequences are formed by addition, **geometric sequences** are formed by multiplication. Each term in a geometric sequence is found by multiplying the previous term by the same number. The multiplier in a geometric sequence is called the **common ratio** denoted by r.

Geometric Sequence

Formula:

$$a_n = a_1 r^{n-1}$$

Where,

a_n= is the last term

a_1= is the first term

n= is the number of terms

r= is the common ratio

Illustrative Example 2:

Sequence: 3, 6, 12, 24, …

Common ratio: $r = 2$

Next three terms: 48, 96, 192

Fibonacci Sequence

is a sequence of numbers in which each term is obtained by adding the two preceding terms. This type of sequence does not have a common difference or a common ratio.

Each term is the *sum of the two previous terms*.

The sequence starts with the numbers 0 and 1 and continues indefinitely.

Fibonacci Sequence

It can be represented by the recursive formula:

$$F_n = F_{n-1} + F_{n-2}$$

Where,

F_n = is the nth term

F_{n-1} = is the term before F_n

F_{n-2} = is the second term before F_n

Fibonacci Sequence

Illustrative Example 3:

Sequence: 1, 1, 2, 3, 5, …

Next three terms: 8, 13, 21

Harmonic Sequence

is a sequence formed by taking the reciprocals of the terms of an arithmetic sequence. This type of sequence does not have a common difference or a common ratio, but it is based on the pattern of reciprocals.

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Harmonic Sequence

If an arithmetic sequence is:

$$a, a + d, a + 2d, \ldots$$

Then the harmonic sequence is:

$$\frac{1}{a}, \frac{1}{a+d}, \frac{1}{a+2d}, \cdots$$

Illustrative Example 4:

Arithmetic sequence: 2,3,4,5, …

Harmonic sequence: $\frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \ldots$

Next three terms: $\frac{1}{6}, \frac{1}{7}, \frac{1}{8}$

Square Numbers (Sequence of Perfect Squares)

The sequence of square numbers is formed when each term is obtained by **squaring the natural numbers**. This sequence does not have a constant difference or a constant ratio.

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Square Numbers (Sequence of Perfect Squares)

It can be represented by the formula:

$$a_n = n^2$$

Where,

a_n= is the nth term

n= the natural number being squared

Illustrative Example 5:

Sequence: 1, 4, 9, 16, …

Next three terms: 25, 36, 49

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Constant Sequence

is a sequence in which every term is the same. This type of sequence can be considered both arithmetic and geometric because the common difference is 0 and the common ratio is 1.

Constant Sequence

It can be represented by the formula:

$$a_n = \boldsymbol{c}$$

Where,

a_n =is the n^{th}term

n =the constant value

Illustrative Example 6:

Sequence: 1, 4, 9, 16, …

Next three terms: 25, 36, 49

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Determining the Rule of a Pattern

To identify the rule, follow these steps:

1. Write down the sequence.
2. Look for differences between consecutive terms. If the difference is constant, it is arithmetic.
3. Look for ratios between consecutive terms. If the ratio is constant, it is geometric.
4. If neither applies, check if it follows Fibonacci or other special rules.

Illustrative Example

If you save ₱50 every week, how would your total savings grow over time?

a) Show patterns in growth (tables, graphs, real data). Discuss what you notice.

b) If the savings continue, how much will you have after 10 weeks? 25 weeks? Can you predict it without listing all terms?

Solution:

a) Show patterns in growth (tables, graphs, real data). Discuss what you notice.

$Week\ (n)$	$Savings\ (a_n)$
1	₱50
2	₱100
3	₱150
4	₱200
5	₱250

Solution:

b) If the savings continue, how much will you have after 10 weeks? 25 weeks? Can you predict it without listing all terms?

10 weeks: ₱500

25 weeks: ₱1,250

Since:

First term: $\boldsymbol{a_1 = 50}$

Common difference: $\boldsymbol{d = 50}$

Use the arithmetic sequence formula:

$$a_n = a_1 + (n - 1)d$$

Substitute the values:

$$a_n = 50 + (n - 1)(50)$$

Simplify:

$$a_n = 50n$$

Let's Check!

For **10 weeks**:

$$a_{10} = 50(10) = ₱500$$

For **25 weeks**:

$$a_{25} = 50(25) = ₱1,250$$

Let's Check!

For **10 weeks**:

$$a_{10} = 50(10) = ₱500$$

For **25 weeks**:

$$a_{25} = 50(25) = ₱1{,}250$$

Illustrative Example

A viral video gets twice as many views each day compared to the previous day. On Day 1, it had 1,000 views.

a) Show patterns in growth (tables, graphs, real data). Discuss what you notice.

b) If the views continue, how many will you have after a week? 10 days? Can you predict it without listing all terms?

Solution:

a) Show patterns in growth (tables, graphs, real data). Discuss what you notice.

$Day\ (n)$	$Views\ (a_{\mathrm{n}})$
1	1,000
2	2,000
3	4,000
4	8,000
5	16,000

Solution:

b) If the views continue, how many will you have after a week? 10 days? Can you predict it without listing all terms?

Day 7: 64,000 views

Day 10: 512,000 views

Solution:

b) If the views continue, how many will you have after a week? 10 days? Can you predict it without listing all terms?

Finding the Rule

Given:

First term: $a_1 = 1,000$

Common ratio: $r = 2$

Use the geometric sequence formula:

$$a_n = a_1(r)^{n-1}$$

Substitute the values:

$$a_n = 1000(2)^{n-1}$$

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Let's Check!

For **7 days**:

$$a_7 = 1000(2)^{7-1}$$
$$a_7 = 1000(2)^{6}$$
$$a_7 = 1000(64) = 64,000$$

For **10 days**:

$$a_{10} = 1000(2)^{10-1}$$
$$a_{10} = 1000(2)^{9}$$
$$a_{10} = 1000(512) = 512,000$$

Illustrative Example

A pair of rabbits is placed in a field. Each month, every pair produces another pair of rabbits starting from the second month.

a) Show patterns in growth (tables, graphs, real data). Discuss what you notice.

b) If the pattern continues, how many rabbit pairs will there be after 6 months? After 10 months? Can you predict the population growth without listing all terms?

Solution:

a) Show patterns in growth (tables, graphs, real data). Discuss what you notice.

$Month\ (n)$	$Pair\ of\ Rabbits\ (a_n)$
1	1
2	1
3	2
4	3
5	5

b) If the pattern continues, how many rabbit pairs will there be after 6 months? After 10 months? Can you predict the population growth without listing all terms?

If the pattern continues after 6 months, the result is

$$a_6 = a_4 + a_5 = 3 + 5 = 8$$

After 10 months, the population will be

$$a_7 = a_5 + a_6 = 5 + 8 = 13$$

$$a_8 = a_6 + a_7 = 8 + 13 = 21$$

$$a_9 = a_7 + a_8 = 13 + 21 = 34$$

$$a_{10} = a_8 + a_9 = 21 + 34 = 55$$